Student Feedback Intelligence

Aspect-Based Sentiment Analysis of Academic Feedback

Week 7: Integration, Demo and Responsible AI

SNLP Course Project 2026 • Group 5

E029 · E030 · E031 · E033 · E034 · E035

Document version 1.0 • Model transformer-absa-0.2.2 • Dataset v1.1-edurabsa
No model weight, threshold or rule was changed this week, and the test split was not opened.

1. Scope and Status

The handout's Week 7 targets, and where each one stands:

Handout target (§14, Week 7)	Status	Where
API integration with another project	**Done on our side.** A course-level signal endpoint for Group 6, and a comment handoff in Group 6's own format. Group 6's reply is due 28 October	Section 2
Integration with the AI University architecture	Done	Section 3
Responsible-AI analysis	**Done.** One-page statement (final deliverable I)	Section 4, `docs/Responsible_AI_Statement.pdf`
Documentation	**Done.** README answers the handout's "reusable asset test"; API guide and specification updated	Section 5
User interface / demo	The `/docs` page and the terminal demos work; one new demo for Group 6. A one-page web interface moves to Week 8	Sections 2.2, 6

All 35 automated tests pass (`results/week7_tests.txt`): the 31 from Week 6, plus four for this week's work.

2. Integration with Group 6 (Early Warning)

The handout names *Group 5 → Group 6: student feedback analysis → student-support signals* as an integration point (§22). We built both directions of that link.

2.1 What Group 6 asked for, and what we send

On 24 October Group 6 proposed the following: we send them feedback comments as `{"items": [{"id": "FB001", "text": "..."}]}`, and they return, for each id, a support-signal category (academic, financial, mental health, harassment, administrative), an urgency, a confidence and a recommended action.

`src/integration_group6.py request` builds that file (`results/week7_group6_request.json`):

	Count
Comments sent	38
From the **train** split (validation and test untouched)	30
of which have at least one negative aspect in the gold labels	15
Hand-written, marked `"synthetic": true`	8
Comments where a titled name was redacted	2

Three privacy rules shaped the sample:

- **No comment that may name a person is sent.** Our redaction misses bare names, so the script skips any comment with a capitalised word mid-sentence, or a capitalised word that never appears in lowercase in the corpus. The first draft of the sample, before this rule, contained instructor names and a remark about an instructor's disability. That is exactly the gap described in Section 4.
- Comments EduRABSA files under *Personal traits* are skipped.
- The rest are redacted with the course-report rule (`pipeline.redact_quote`).

The public reviews rarely mention money, health or harassment. The 8 synthetic comments therefore exercise Group 6's categories: two are the handout's and Group 6's own examples. A second file, `integration/group6_request_with_analysis.json`, adds our analysis (document sentiment, aspects and polarity) to every item, so Group 6 can use our output, not just the raw text.

`src/integration_group6.py merge <reply>` reads Group 6's reply back. It checks every id against what we sent and writes **counts only** to `results/`: flags per category and urgency, and which of our aspects occur in flagged comments. No comment text or id appears in the summary.

Until Group 6's reply arrives, a stand-in is used and labelled as one. `integration_group6.py mock` is a keyword stand-in written by us. On it, the merge step read 38 of 38 ids back with no unknown or missing id

(`results/week7_group6_merge_mock.json`). This tests our plumbing only; it says nothing about Group 6's system. The real reply will be merged the same way and reported in Week 8.

One observation from the mock run concerns our own model: it tags some synthetic non-academic comments (a part-time job, a harassment report) with *teaching: negative*. This is the teaching-negative bias from Week 5, seen again in a new setting.

2.2 What Group 6 receives from our API: course-level signals

The contract designed in Week 1–2 (API specification §8) is now served at `POST /v1/support-signals`. The request is the same as for the course report. The response holds one `academic_concern` signal per aspect that the course report already lists as a **top concern** or an **emerging concern**, so no new threshold was invented. On the 150-comment demo course (`results/week7_support_signals_demo.txt`):

Aspect	Basis	Negative mentions	Share of comments
workload	top concern	12 of 150	8.0%
pace	emerging concern	3 of 150	2.0%

Each signal carries `advisory_only: true` and `not_for_individual_action: true`. No quote, comment or id is ever included, and a test enforces this. The demo comments are the same 150 test-split comments used for the Week 6 course report demo; no metric is computed on them.

2.3 What we need from Group 6

- Their reply to the 38 comments by **28 October**.
- One example of their response format now: field names, the list of categories and urgency levels, and the confidence range.
- A choice for the final demo: a file handoff, or a live endpoint.

3. Place in the AI University Architecture

Layer (handout §4)	Our part
University data	Anonymised course feedback (here: public EduRABSA reviews)
Shared NLP services	Text classification and information extraction: aspect-based sentiment with evidence spans and calibrated scores
Student services → Feedback	Course reports for coordinators: `POST /v1/analyze-feedback/batch`
Student services → Early Warning (Group 6)	Course-level concern signals (`POST /v1/support-signals`) and the comment handoff (Section 2.1)

4. Responsible AI

The one-page statement (`docs/Responsible_AI_Statement.pdf`) covers privacy, bias, hallucination, human oversight, security and limitations. Its main points:

- **Privacy.** Public, licensed data only; local model; PII redaction. The known gap is names without a title. This week it became concrete: it is why the Group 6 sample excludes any comment that may contain a name.
- **Bias.** In 5 of the 15 failed behaviour probes, the error is the same spurious *teaching: negative* (`results/week5_probe_results.json`), which is unfair to instructors. Hinglish probes pass 0 of 5 and sarcasm 0 of 4 (`results/week5_eval.json`).
- **Hallucination.** The model is extractive and generates no text; quotes are verbatim.
- **Human oversight.** Calibrated scores, `low_confidence` warnings, small-sample caveats, and flags that forbid individual action.

5. Documentation

- `README.md` now opens with the handout's reusable-asset test (§21): input, output, interface, an example, how to run it, how well it works, and what it must not be used for.
- `docs/API.md` has a section for `POST /v1/support-signals`; `docs/API_Specification.md` goes to v2.2.
- The new code is documented in its module docstring (`src/integration_group6.py`), with tests in `tests/test_integration.py` and `tests/test_api.py`.

6. Moved to Week 8, and Why

Week 7 was submitted late, so these items move to Week 8. None changes the model or any reported number:

Item	Week 8 plan
One-page web interface	A single page served by the API: paste comments, see aspects and the course report
Topic clustering (handout: “topic coherence or cluster quality”)	Diagnostic on the train split only, as in `docs/Evaluation_Protocol.md` Task D
Group 6's real reply	Merge it, report the counts, show the handoff in the demo
Architecture diagram	Redrawn for the final system
Final report, slides, demo script, Git repository	Final submission, 31 October

Appendix A. Reproducing every number

```
python -m pytest tests -q                                  # results/week7_tests.txt
python src/integration_group6.py request                   # results/week7_group6_request.json
python src/integration_group6.py mock
python src/integration_group6.py merge integration/group6_response_MOCK.json
python src/api.py          # then, in a second window:
python examples/demo_support_signals.py                    # results/week7_support_signals_demo.txt
```